1 (v) $E[X]=E[X_B]-E[X_R]=5\cdot\frac{13}{21}-5\cdot\frac{8}{21}=\frac{25}{21}$

method 1:

$X_B$ and $X_R$ not independent

$X_B+X_R=5$

$X=2X_B-5$

$Var(X)=Var(2X_B-5)=2^2\,Var(X_B)=4\cdot5\cdot\frac{13}{21}(1-\frac{13}{21})=\frac{2080}{441}$

method 2: or $Var(X_B-X_R)=Var(X_B)+Var(X_R)-2Cov(X_B,X_R)$

$Cov(X_B,X_R)=Cov(X_B,5-X_B)=Cov(X_B,-X_B)$

$=-Var(X_B)$

$Var(X_B-X_R)=3Var(X_B)+Var(X_R)$

$=3\cdot5\cdot\frac{13}{21}(1-\frac{13}{21})+5\cdot\frac{8}{21}(1-\frac{8}{21})$

$=4\cdot5\cdot\frac{13}{21}\cdot\frac{8}{21}=\frac{2080}{441}$

Properties of $Cov(X,Y)$

① $Cov(X,cY)=c\,Cov(X,Y)$

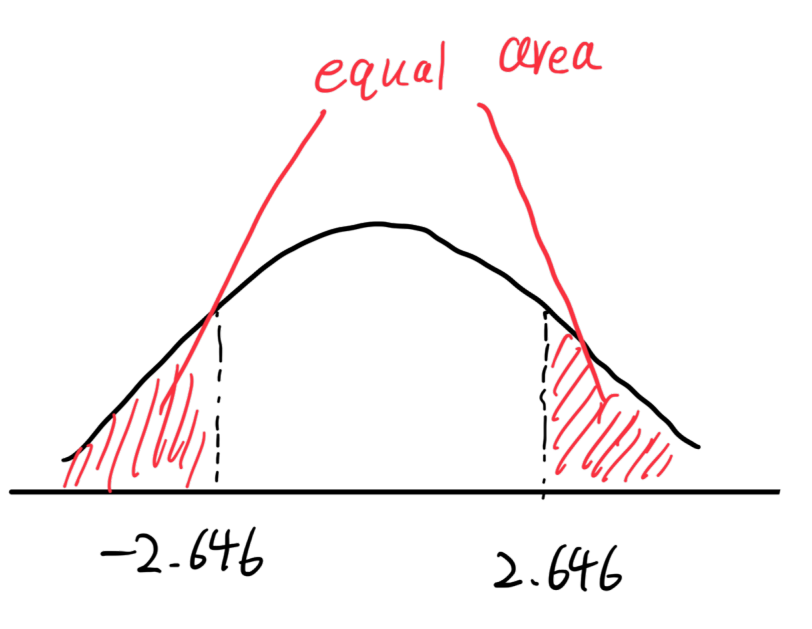
② $Cov(X,a)=0$, where $a$ is constant

③ $Cov(X,Y+Z)=Cov(X,Y)+Cov(X,Z)$

④ $Cov(X,X)=Var(X)$

1 (vi) $P(Z\geq2.646)$

cdf $\Phi(-2.646)$



4. (vi) $\begin{cases} Z=1 & X \text{ is even} \\ Z=0 & X \text{ is not even} \end{cases}$

independent: $P(Z=1\mid N=n)=P(Z=1)$

for $b(n,p)$, $P(X\text{ is even})=\sum_{k\text{ even}}\binom{n}{k}p^k(1-p)^{n-k}$, denoted $P(\text{even})$

$P(X\text{ is odd})=\sum_{k\text{ odd}}\binom{n}{k}p^k(1-p)^{n-k}$, denoted $P(\text{odd})$

① $P(\text{even})+P(\text{odd})=1$

② $P(\text{even})-P(\text{odd})=\sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}\cdot(-1)^k=\sum_{k=0}^{n}\binom{n}{k}(-p)^k(1-p)^{n-k}$

$$\begin{cases} ①\ [p+(1-p)]^n=\sum_{k=0}^{n}\binom{n}{k}p^k(1-p)^{n-k}=1 \\ ②\ [(1-p)-p]^n=\sum_{k=0}^{n}\binom{n}{k}(-p)^k(1-p)^{n-k}=(1-2p)^n \end{cases}$$

$$\begin{cases} P(\text{even})+P(\text{odd})=1 \\ P(\text{even})-P(\text{odd})=(1-2p)^n \end{cases}$$

$P(\text{even})=\frac{1+(1-2p)^n}{2}$, which should be independent with n

so $p=0$ or $p=\frac{1}{2}$

In EXAM, you can try values of P one by one!

6. (v) $Cov(XY)=E[XY]-E[X]E[Y]$

$E[Y]=\int_0^2 y f_Y(y)\,dy$

$f_Y(y)=\int_0^{2-y}\frac{3}{2}xy\,dx=\frac{3}{4}y(2-y)^2$

$E[Y]=\int_0^2\frac{3}{4}y^2(2-y)^2dy=E[X]=\frac{4}{5}$

$E[XY]=\int_0^2\int_0^{2-x}f(x,y)\cdot xy\,dy\,dx=\frac{3}{2}\int_0^2\int_0^{2-x}x^2y^2dy\,dx=\frac{8}{15}$

$Cov(X,Y)=\frac{8}{15}-(\frac{4}{5})^2=-\frac{8}{75}$

Here $\theta\leq\min(X_1,X_2,\dots,X_n)$

7 (v) $L(\theta)=\prod_{i=1}^{n}\frac{3}{(X_i-\theta+1)^4}\mathbb{1}_{[\theta,+\infty)}(X_i)=3^n\prod_{i=1}^{n}(X_i-\theta+1)^{-4}$

$\ln L(\theta)=n\ln3-4\sum_{i=1}^{n}\ln(X_i-\theta+1)$

$\frac{\partial}{\partial\theta}\ln L(\theta)=-4\cdot(-1)\sum_{i=1}^{n}\frac{1}{X_i-\theta+1}=4\sum_{i=1}^{n}\frac{1}{X_i-\theta+1}>0$

$\theta\uparrow$, $\ln L(\theta)\uparrow$, $L(\theta)\uparrow$

$\hat{\theta}_{MLE}=\min(X_1,X_2,\dots,X_n)$

for each $X_i$, $X_i>\theta$

$E[\hat{\theta}_{MLE}]>\theta$

$\hat{\theta}_{MLE}$ biased

8 (v) $L(\theta)=\frac{1}{(2\delta)^n}$, where $T-\delta\leq X_i\leq T+\delta$

To max $L(\theta)$, $\delta=\delta_{min}$, $\begin{cases} T-\delta\leq\min(X_1,X_2,\dots,X_n) \\ T+\delta\geq\max(X_1,X_2\dots X_n) \end{cases}$

$\begin{cases} \delta\geq T-X_{min} \\ \delta\geq X_{max}-T \end{cases}$ $\delta\geq\max\{T-X_{min},X_{max}-T\}$

① when $T<\frac{1}{2}(X_{min}+X_{max})$, $X_{max}-T>T-X_{min}$

$\delta=X_{max}-T>\frac{1}{2}(X_{max}-X_{min})$

② when $T>\frac{1}{2}(X_{min}+X_{max})$, $T-X_{min}>X_{max}-T$

$\delta=T-X_{min}>\frac{1}{2}(X_{max}-X_{min})$

③ when $T=\frac{1}{2}(X_{min}+X_{max})$, $T-X_{min}=X_{max}-T$

$\delta=\frac{1}{2}(X_{max}-X_{min})$

So $\delta$ get $\delta_{min}$ when $T=\frac{1}{2}(X_{min}+X_{max})$

$\delta_{min}=\frac{1}{2}(X_{max}-X_{min})$

1. Organize Your Draft Paper: Fold your draft paper to create several sections for each problem. This will save your time when you check your calculation process.

2. Master Your materials : For this open-book exam, be familiar with the position of formulas and examples in your materials. Methods to new questions are same as examples you've already learned.

3. Connect the Dots Within a Question: For multi-part questions, pay close attention to previous small questions. The results of these questions may help you solve next questions.